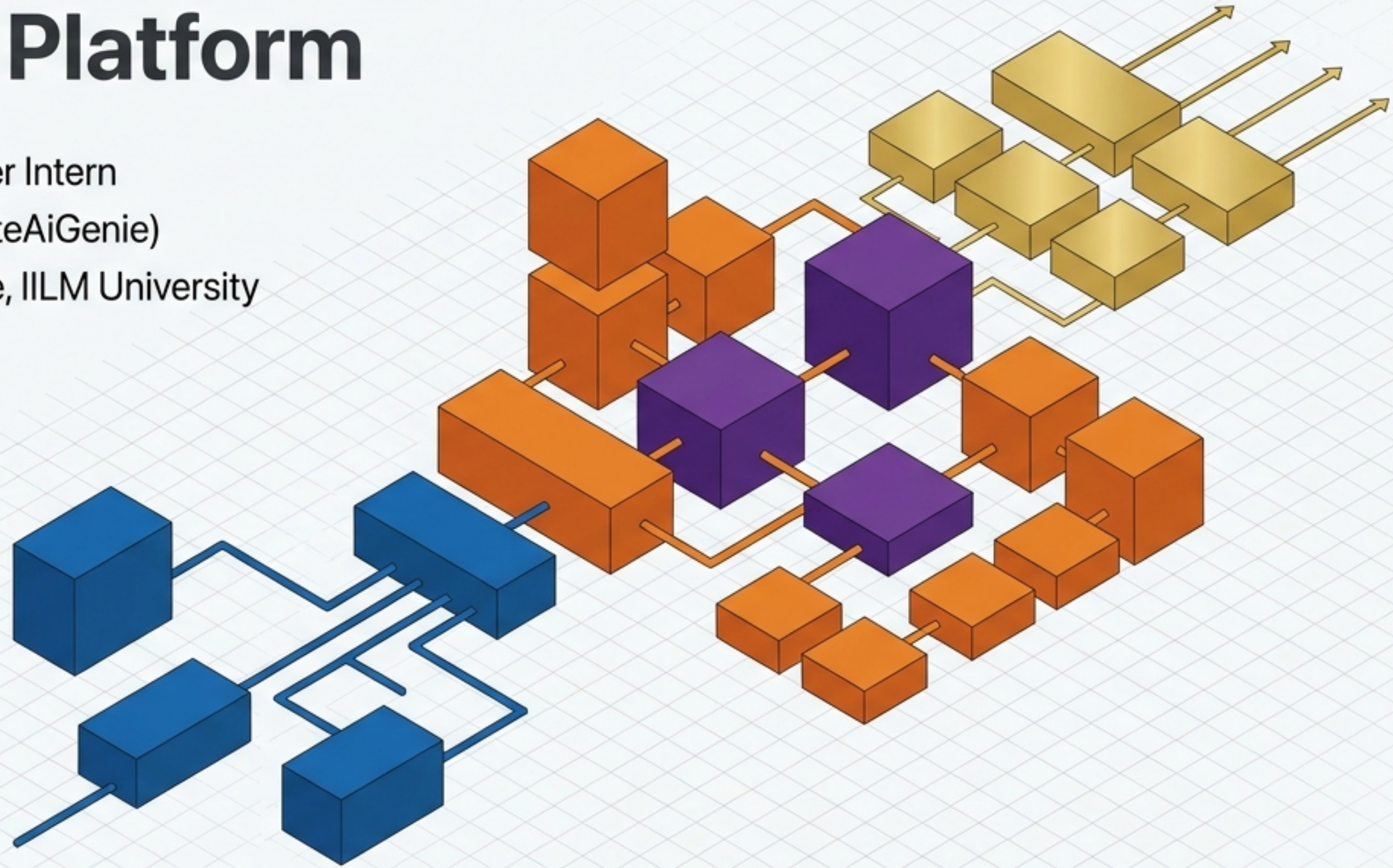
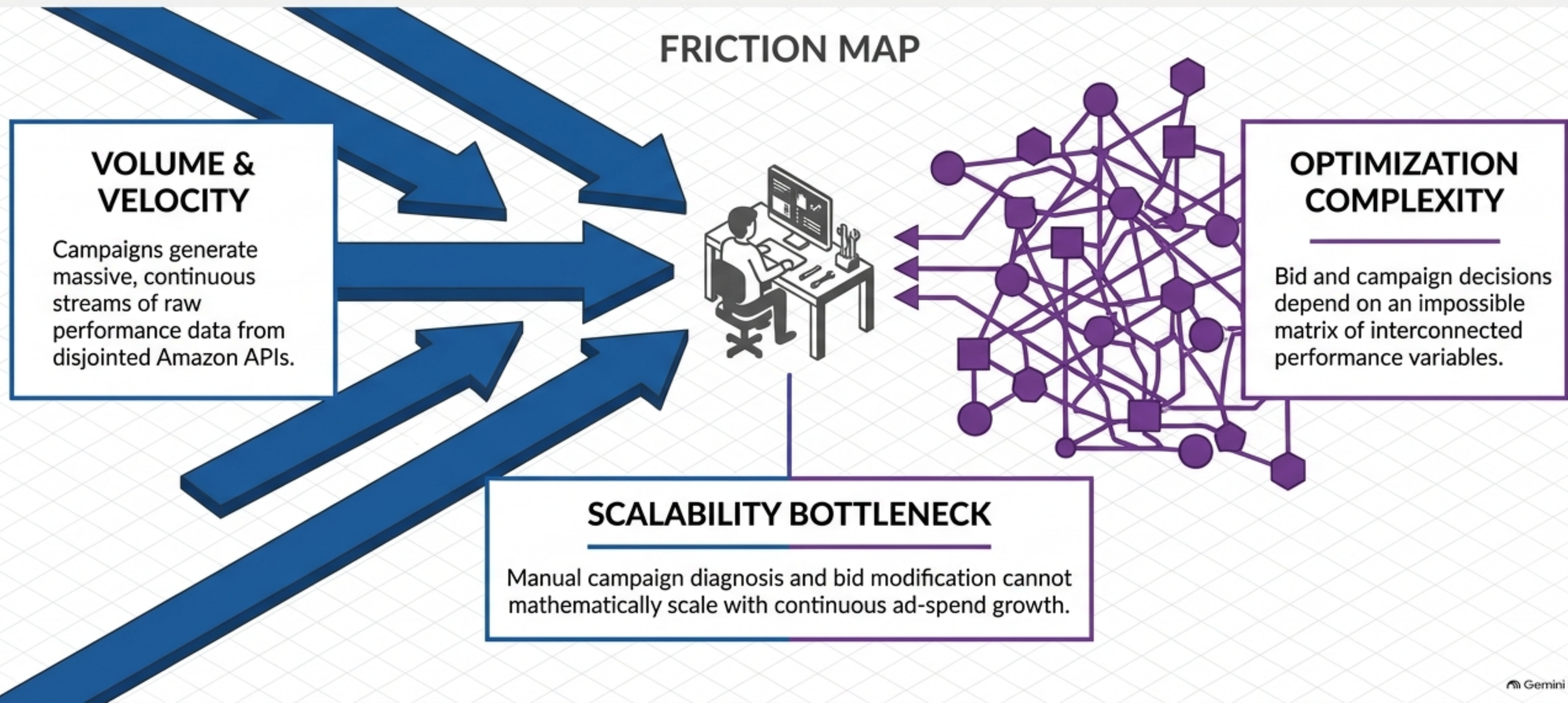


AI-Driven Amazon Ads Optimization Platform

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April 2026 - Completion



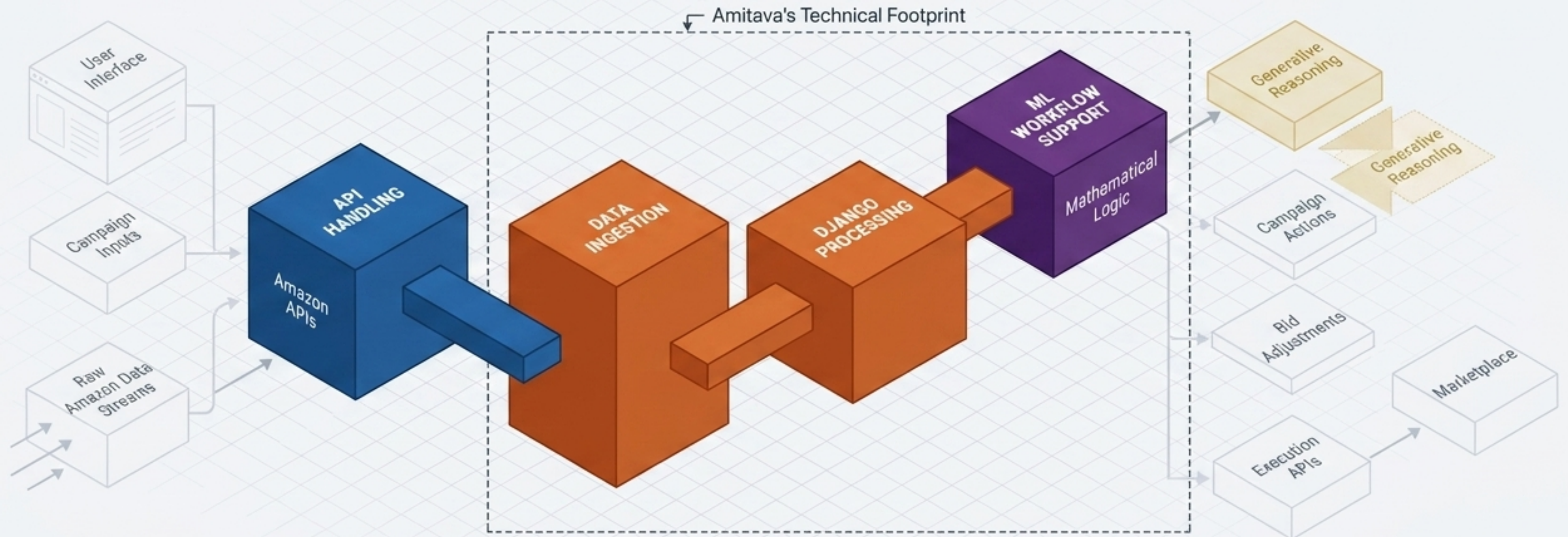
MANUAL OPTIMIZATION CRUMBLES UNDER HIGH-VELOCITY DATA



ARCHITECTURAL SCOPE AND INTERNSHIP FOOTPRINT

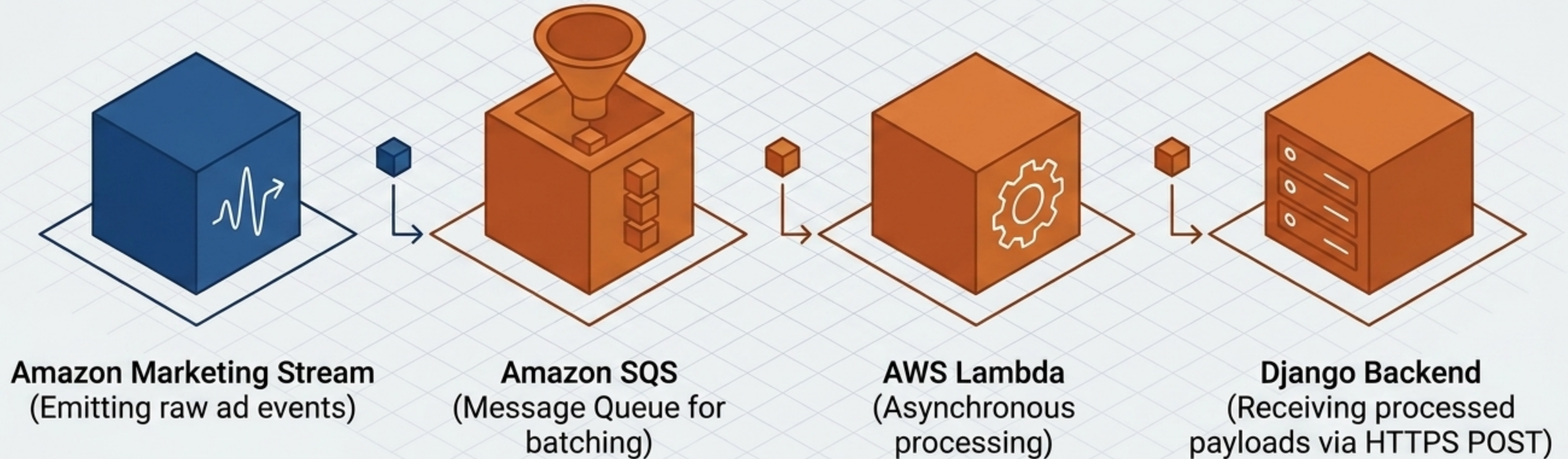
THE SYSTEM: An enterprise platform converting raw Amazon data into intelligent campaign actions.

THE CONTRIBUTION: Designing the API data-flow, establishing ingestion pipelines, and structuring raw data for ML and generative reasoning layers.



Ingesting Asynchronous Data Streams

Data arrives continuously; direct REST integration would fail under load. SQS and Lambda ensure reliable, batched, asynchronous handoffs to the backend.

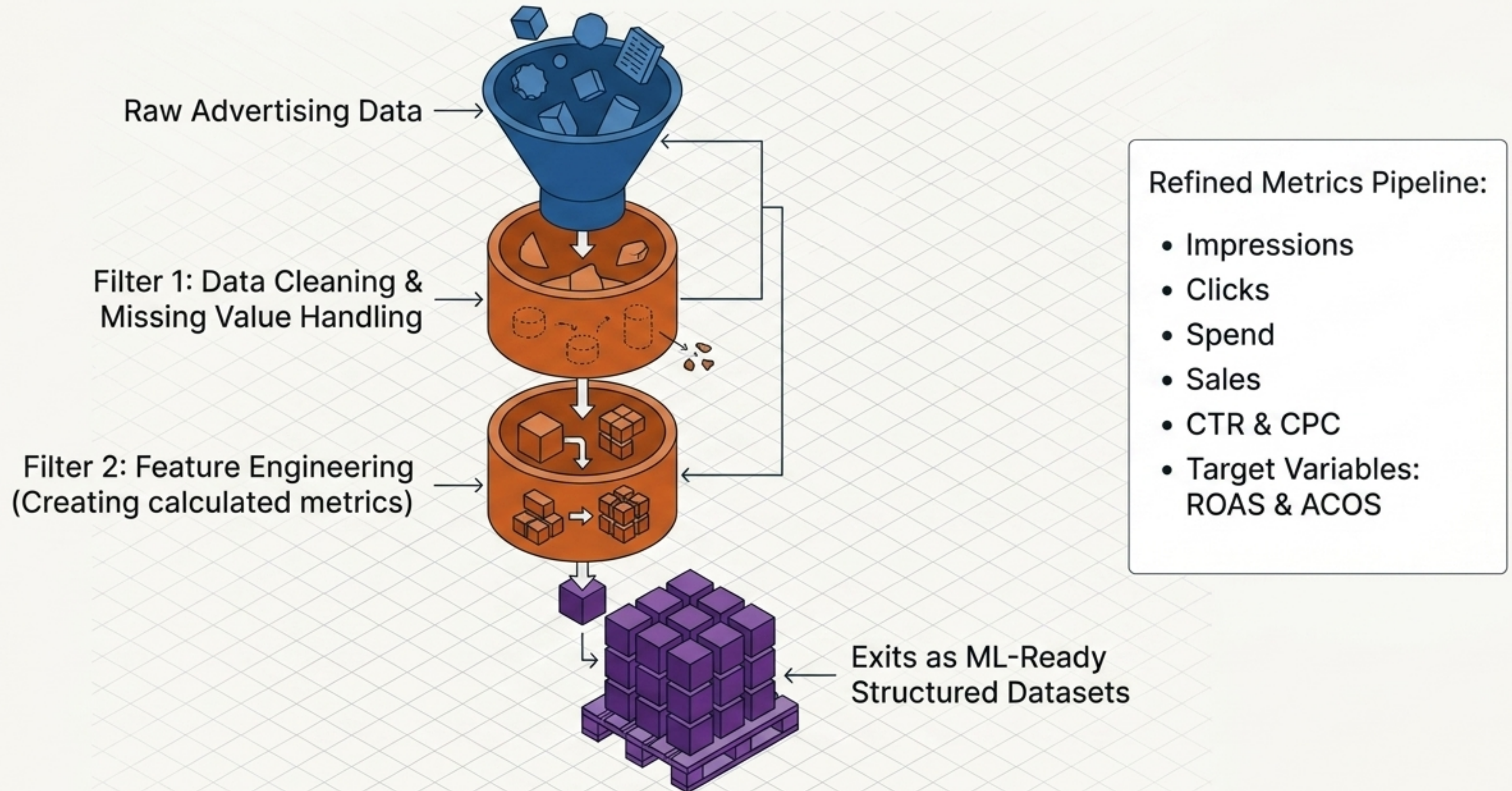


Context-Specific API Security Protocols

Different communication paths demand distinct security models. IAM controls the underlying AWS resource permissions throughout.



Refining Raw Metrics for Machine Learning



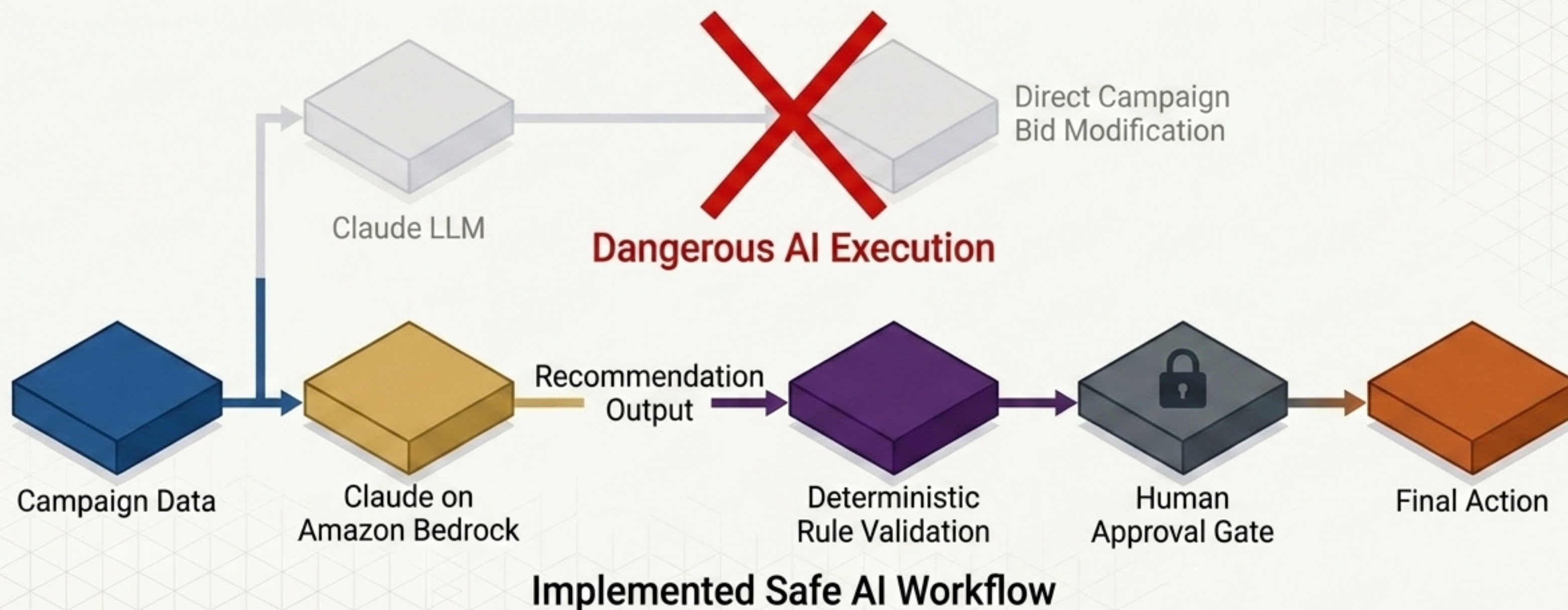
Decision Matrix: Deterministic Rules vs. Predictive ML

The platform balances dual engines: rules for immediate, predictable safety boundaries, and ML for complex performance forecasting.

Dimension	Rule-Based Logic	Machine Learning Approach
Logic Type	Deterministic (IF/THEN)	Predictive & Probabilistic
Trigger	Explicit threshold (e.g., ACOS > 20%)	Historical pattern recognition
Complexity Handling	Fails under multi-variable friction	Excels at complex, non-linear relationships
Validation	Easy to audit and trace	Requires statistical validation

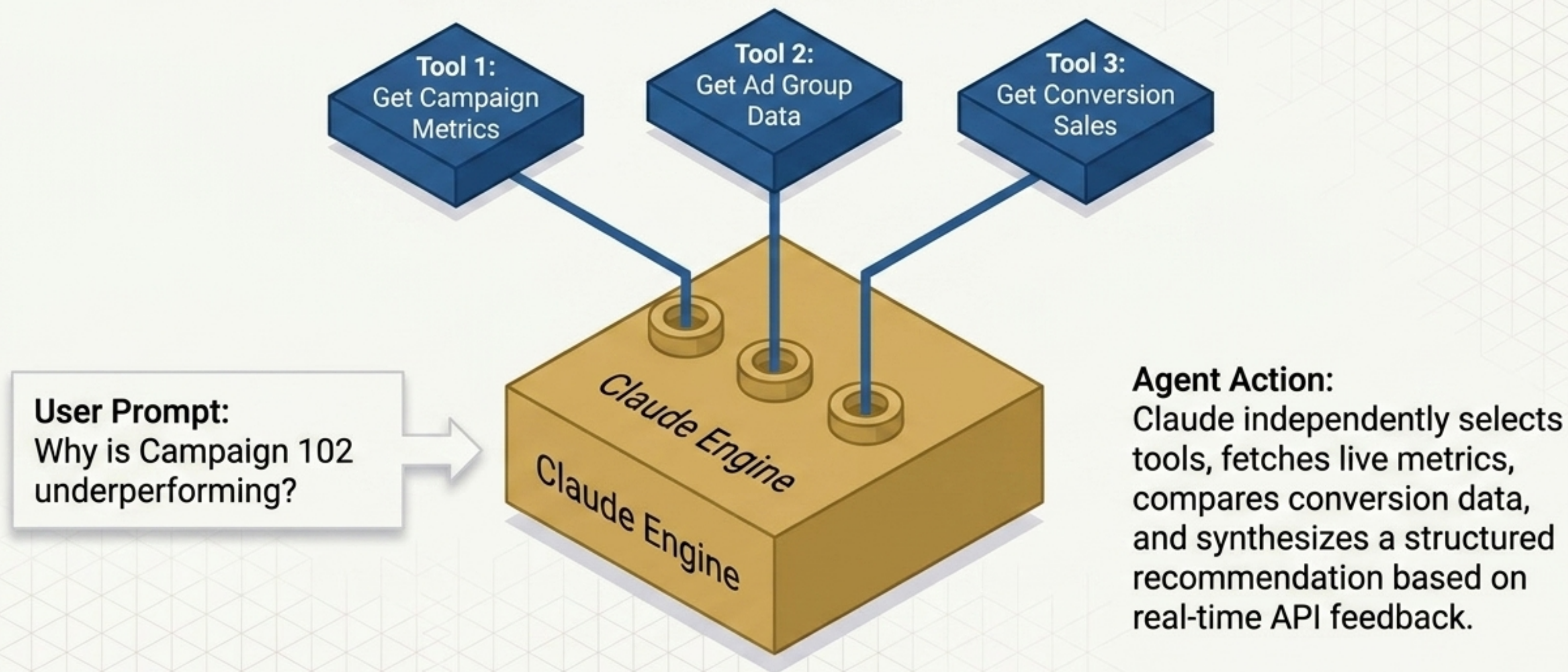
Isolating Generative AI Behind Validation Layers

Claude acts purely as a reasoning layer for diagnosis and strategy generation. Business-critical execution remains strictly gated behind deterministic application logic and human oversight.



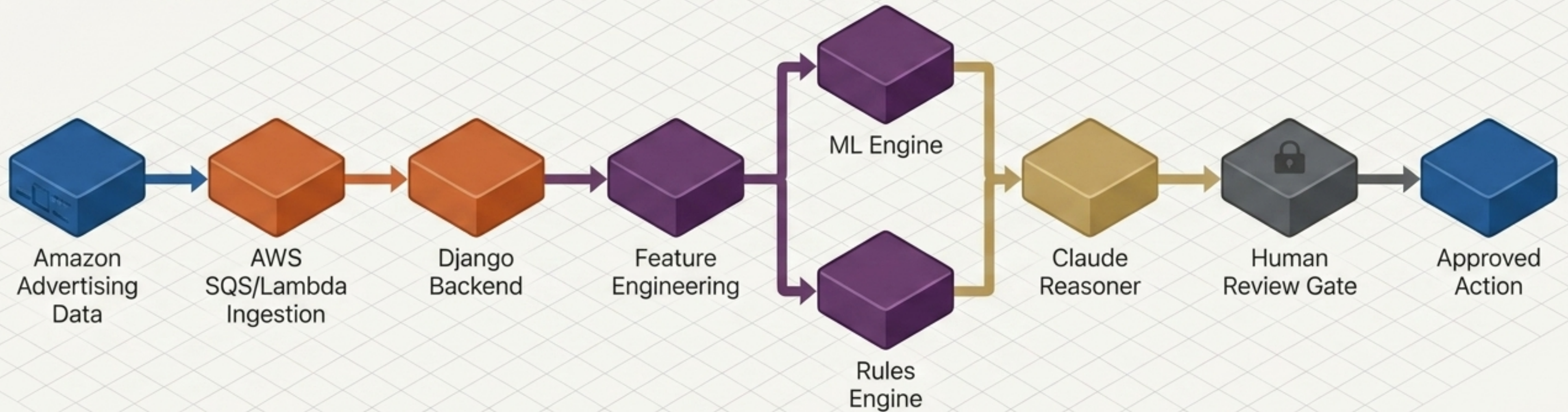
Agentic Data Retrieval for Diagnostic Reasoning

An isometric visual representing an "Agentic Tool-Use Architecture"



The End-to-End Data Lifecycle

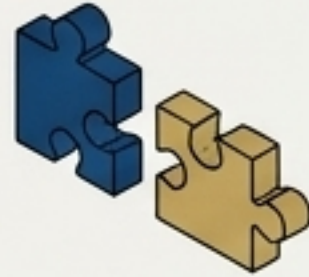
The architecture proves that an intelligent AI system requires reliable data pipelines long before the model is ever invoked.



Engineering Frictions and System Resolutions

Friction: Divergent API mechanisms.

Resolution: Standardized request/response lifecycles across external services.



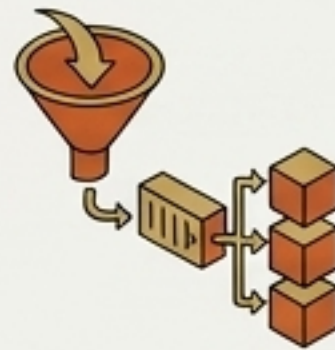
Friction: Mixed security contexts.

Resolution: Mapped JWT, OAuth, and HMAC precisely to user, external, and internal trust boundaries.



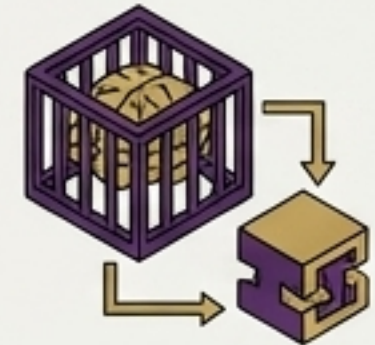
Friction: Continuous data volume.

Resolution: Implemented asynchronous SQS queues to decouple ingestion from processing.



Friction: LLM unreliability.

Resolution: Relegated AI to reasoning; retained execution in deterministic logic.

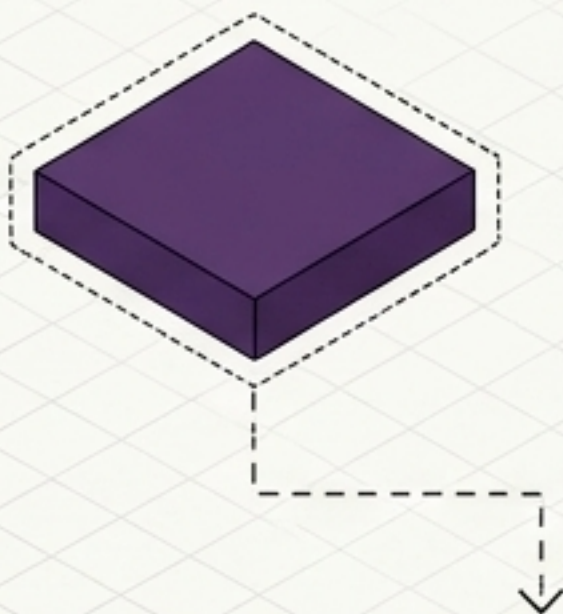


Architectural Evolution and Roadmap

Expanding the platform's autonomy while rigorously maintaining the human-in-the-loop safety gates.

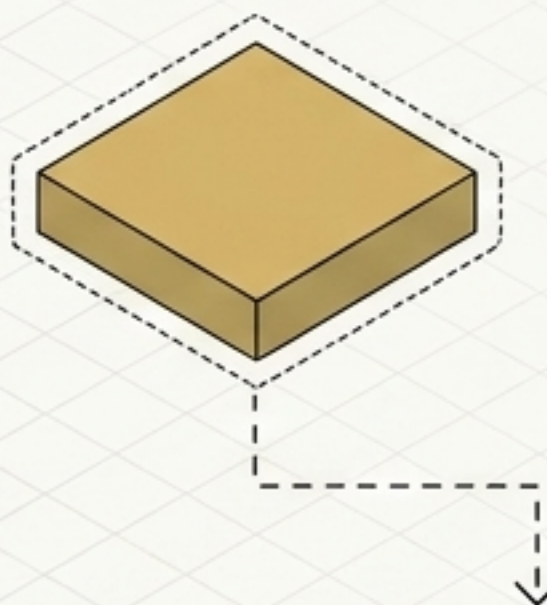
Predictive Optimization

ML models for conversion probability and optimal bid ranges.



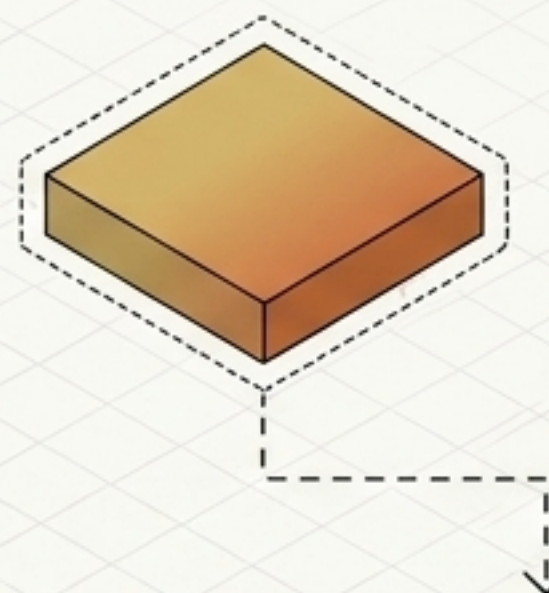
Automated Anomaly Triggers

Continuous monitoring that auto-spins up Claude diagnosis only when thresholds break.



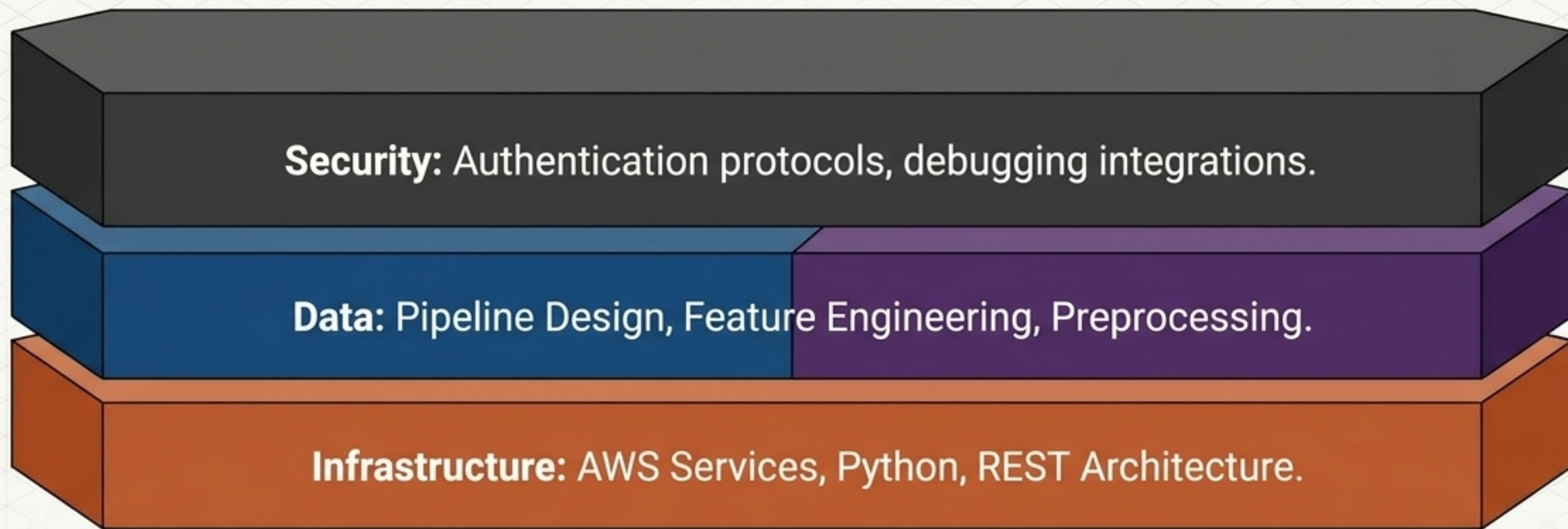
Cost-Aware AI Routing

Dynamic model selection to optimize API spend (smaller models for checks, larger for complex reasoning).



Existing System Framework

Engineering Intelligence Requires Infrastructure



Building an intelligent AI system is not merely about the model—it relies fundamentally on reliable APIs, clean data, secure infrastructure, and well-designed system architecture.